

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)			oxidation state of chlorine goes from -1 to 0 so it is oxidised / oxidation state of nitrogen goes from $+5$ to $+3$ so it is reduced		1		1		
	(b)			enthalpy of formation of elements in their standard state is zero	1			1		
	(c)			$53 + 0 + [2 \times (-286)] - (-173) - [3 \times \Delta_f H^\theta] = -69$ (1) $[3 \times \Delta_f H^\theta] = -277$ $\Delta_f H^\theta = -92.3$ (1)		2		2	2	
	(d)	(i)		$\Delta S^\theta = -90$ (1) convert 25°C to 298K (1) $\Delta G^\theta = -69 - 298 \times (-90) \div 1000$ $= -42.18$ (1)		3		3	2	
		(ii)		the entropy of concentrated hydrochloric acid is lower, so ΔS^θ will be more positive therefore ΔG^θ will be more negative			1	1		
	(e)	(i)		$[\text{H}^+] = 10^{-0.3} = 0.50$		1		1	1	
		(ii)		titration gives the more precise value (1) MUST GIVE REASON TO GAIN THIS MARK pH measurements to fewer significant figures / pH gives wide range of possible concentrations ($0.45 - 0.56$) (1)			2	2		2
				Question 10 total	1	7	3	11	5	2